

Stresses, Strains, and Deflections in Flexible Pavements

Mofreh Saleh, Ph.D., P.E., F.ASCE
University of Canterbury

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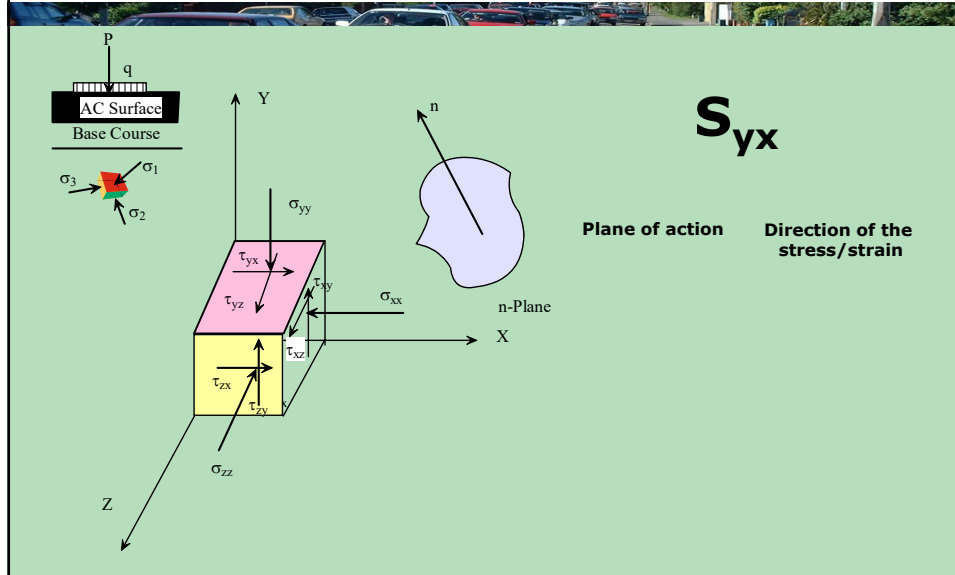
Stresses and Strains in Flexible Pavements



- Solutions for stresses strains in flexible pavements are based on the work done by the French Mathematician Boussinesq. However, this work has its won limitations because it only considers one layer extended laterally.
- Assumptions:
 - 1. Homogeneous infinite half space
 - 2. Elastic and isotropic material

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Stresses and Strains Notations

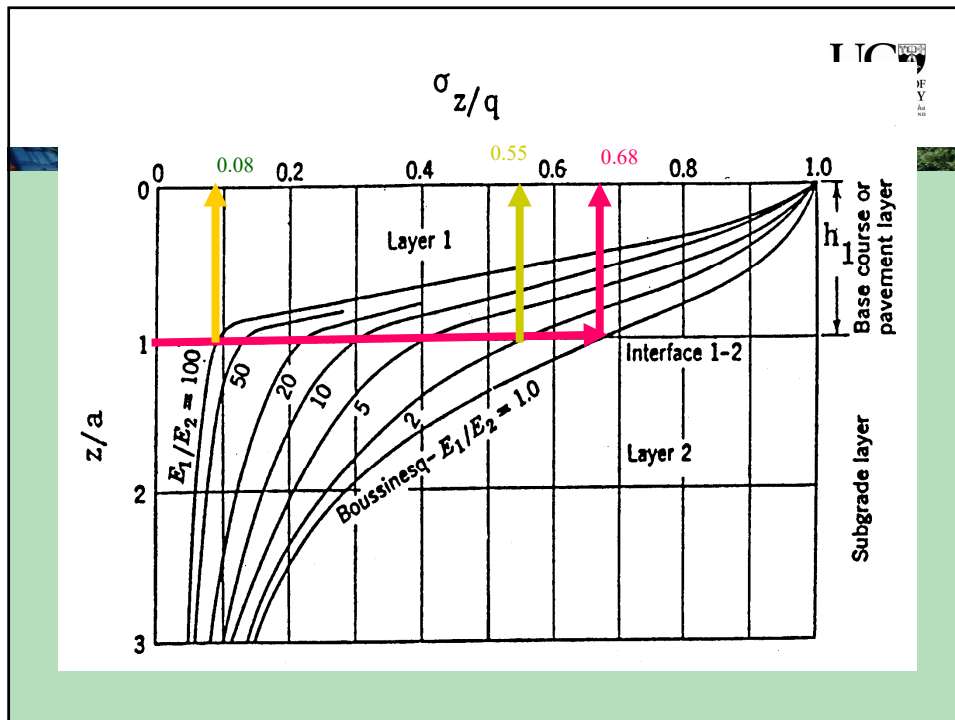


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Effect of Layers Moduli on Interface Stress

- The vertical stresses depends on the ratio of the layers moduli.
- For $E_1/E_2 = 1$ the vertical stresses at the interface is about 68% of the tyre pressure.
- For $E_1/E_2 = 100$, the vertical stress at the interface reduced dramatically to about 8%. This is expected because using stiff and strong pavement will dissipate the stresses significantly at the top of subgrade.

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- Manual solutions for stresses and strains for multiple layers with combined axle loads is quite tedious and complex.
- Numerical solutions offered by computer packages such as Circly, Everstress, BISAR,.. and other packages provide an efficient solution for most pavement design and analysis problems

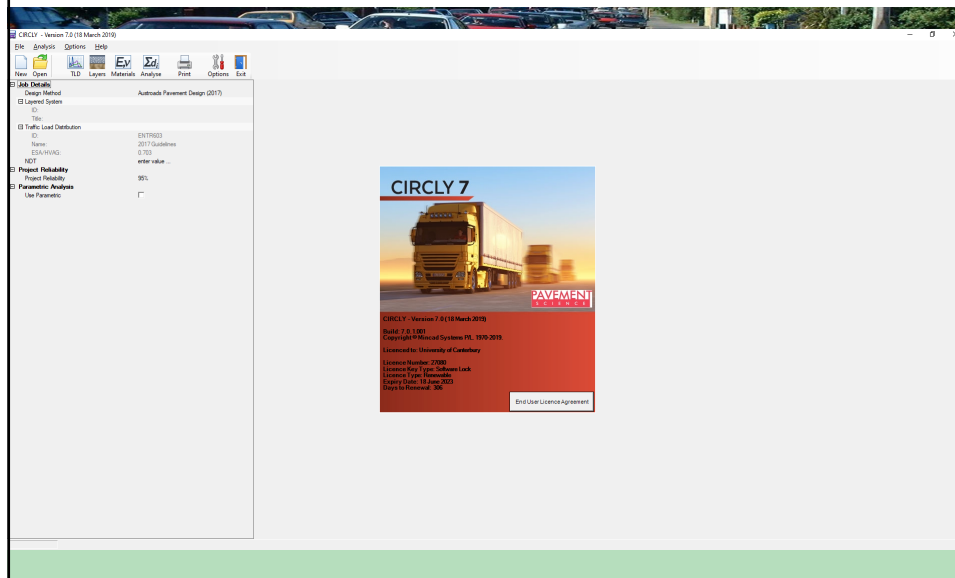
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Circly Software

- Circly offers both stresses, strains, deflections analysis and damage analysis for flexible pavement design.
- In Circly, the user needs to provide complete list of input which includes both Circly functions even if the user intention is to run only stresses, strains, and deflections analysis otherwise the software will always give an error message.
- For Example, if the purpose is to determine the strains at the bottom of the asphalt, user still needs to populate traffic volume otherwise an error message will be displayed.

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Circly 7



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Recommended units of measurement

Quantity	Units
Length, Displacement	mm
Modulus, Pressure	MPa
Force	N
Moment	N.mm
Strain	mm/mm

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Database approach

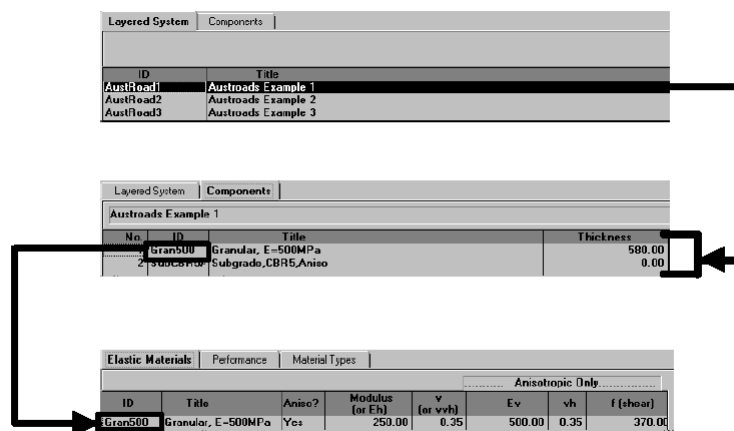
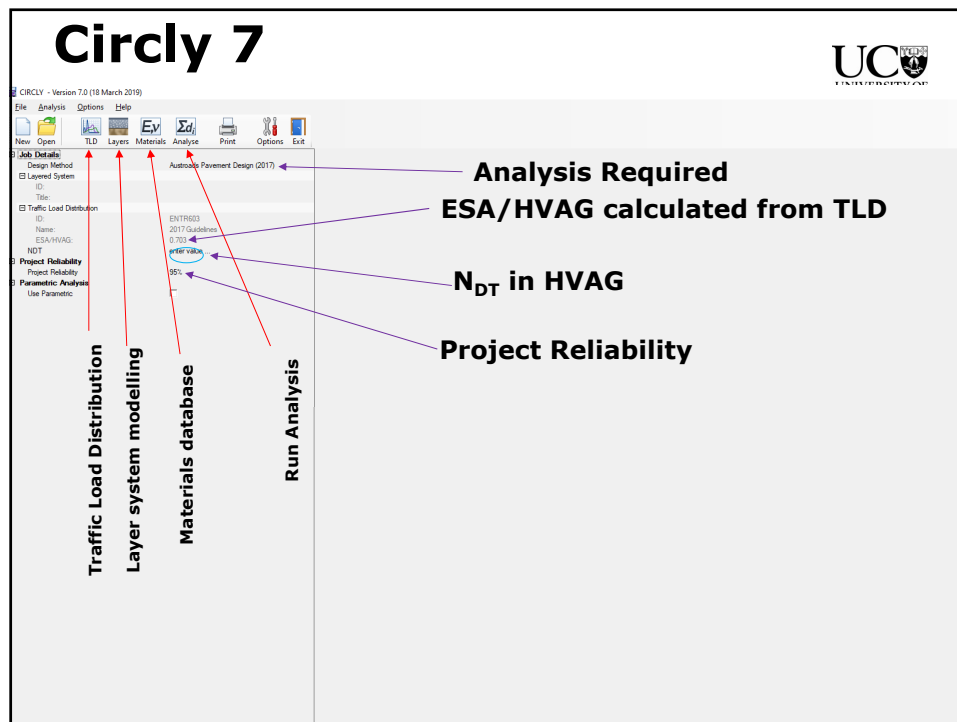
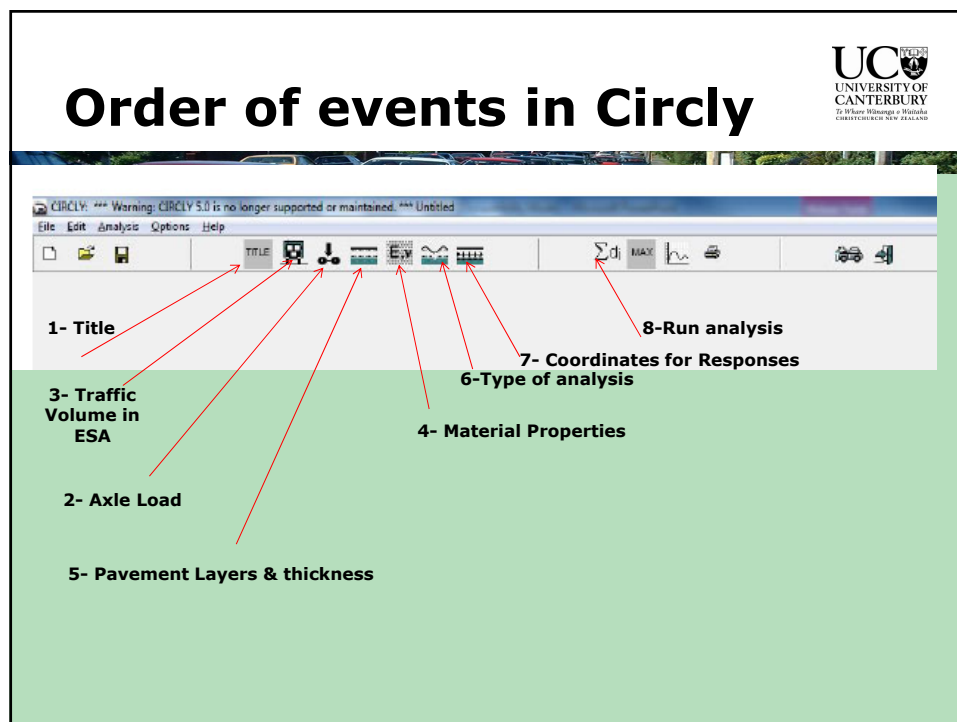


Figure 6 Relationships between elements in Layered System databases

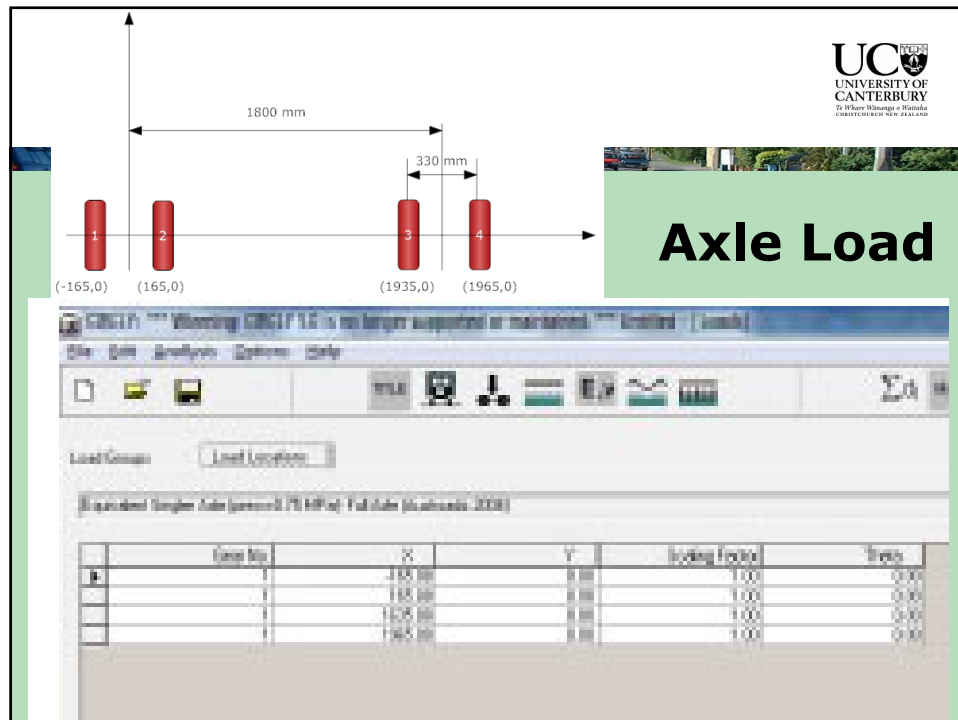
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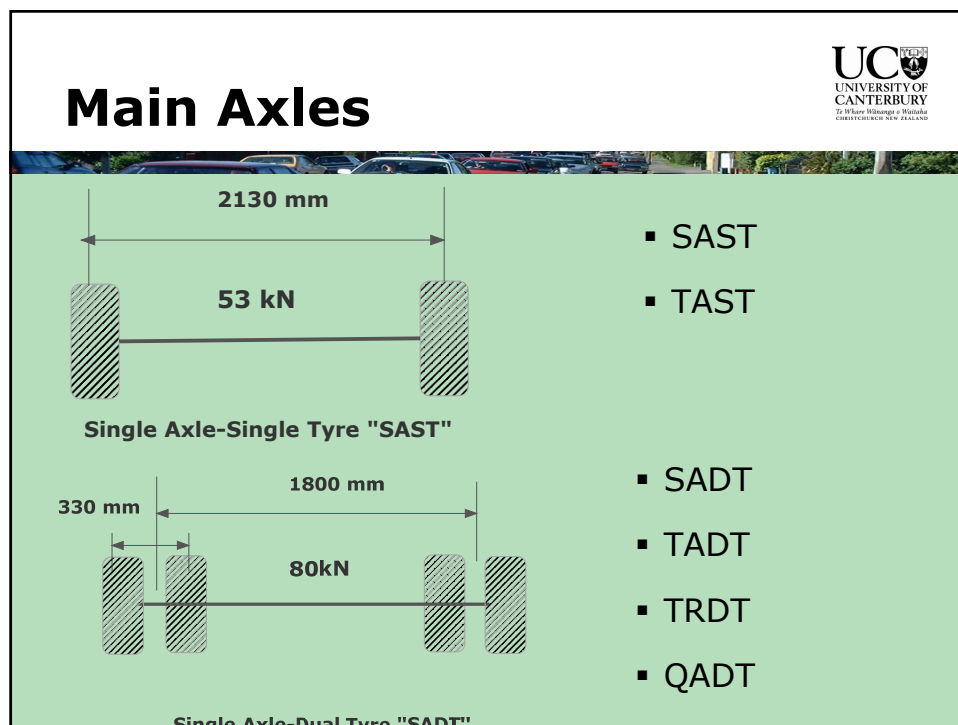
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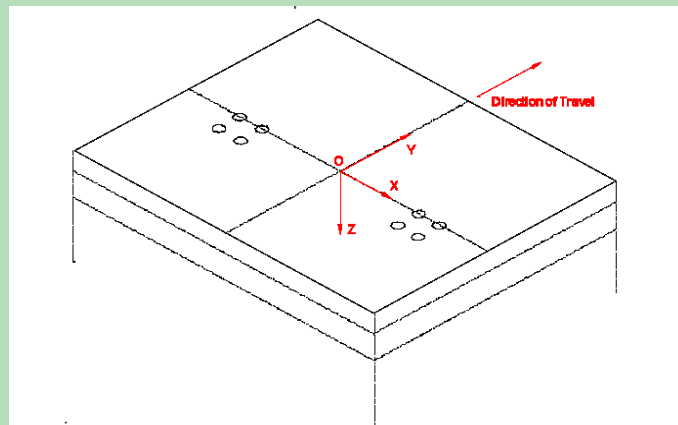


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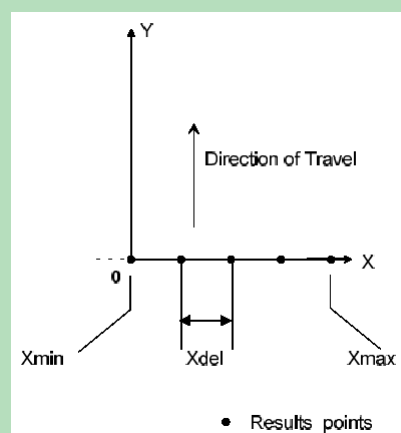
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Global coordinates



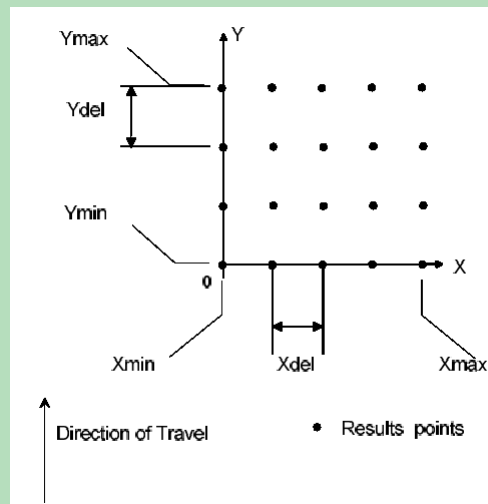
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Specifying analysis points



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Specifying outputs in both x and y directions



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Anisotropic materials

- Austroads design Guide recommends using anisotropic material characterisation for subgrade and granular materials.
- Anisotropic factor of 0.5, ie $E_h = 0.5 E_v$
- Not perfect, but better than isotropic materials.

$$E_h = 0.5 E_v$$

$$\nu_{vh} = \nu_h = \nu$$

$$f = E_v / (1 + \nu)$$

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Example

- If you would like to enter anisotropic basecourse properties for a base with $E_v=400$ Mpa and $\nu=0.35$
- $E_h=200$ MPa
- $\nu_h=\nu_v=0.35$
- $E_v=400$ MPa
- $F=400/(1+0.35)=296.3$
- For isotropic only E_v and ν_h will be required.

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Performance criteria

$$N = \left[\frac{k}{\varepsilon} \right]^b$$

where

k

b

N is the predicted life (repetitions)

k is a material constant

b is the damage exponent of the material

This part is only required for damage analysis in which performance criteria for fatigue and rutting must be established

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Subgrade Strain Criterion

$$N_f = \left(\frac{9150}{\varepsilon_{zz}} \right)^7$$

ε_{zz} = **Compressive strain on the top of the subgrade in microstrain units**

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Performance Criteria for Subgrade

- In Circlly
- Constant must be constant with strain units which is mm/mm, therefore $k = 0.00915$
- Exponent= $b = 7.0$
- Strain= Compressive strain
- Location= on the Top of subgrade

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Asphalt Fatigue Performance Criterion

$$N_{ij} = \frac{SF}{RF} \left(\frac{6918 * (0.856 * V_b + 1.08)}{E^{0.36} * \mu\epsilon_{ij}} \right)^5$$

V_b= percentage by volume of binder usually ranges from 10.5 for dry mix up to 13 for rich mix

E= flexural modulus of the mix (MPa), usually taken as the resilient modulus

μϵ= Tensile strain at the bottom of asphalt in microstrains

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Asphalt Performance

- Example
- Assume asphalt has E=3000 MPa and V_b=12%
- $K=4398.345/10^6=0.004398$
- Exponent=b=5.0
- Type of Strain= horizontal tensile strain
- Location= bottom of the asphalt

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Cement Stabilised Performance Criterion

$$N_{ij} = RF * \left(\frac{K}{\mu \varepsilon_{ij}} \right)^{12}$$

E=Concrete Modulus (MPa)
 $\mu \varepsilon$ =Tensile strain at the bottom of the CTB

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Example

- Create layer properties in Circly
- Create their performance Criteria
- Create Standard Axle Load and Traffic Spectrum
- Create pavement model
- Determine
- Max Surface deflection, maximum tensile strain at the bottom of asphalt and the max compressive strain at the top of the subgrade.

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Circly Analysis

